

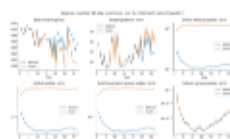
AdamW versus Muon: DistilBERT on SST-2

1. Which optimizer performed better, and by what criterion? Criterion: validation accuracy and loss after 3 matched epochs on SST-2 (67,349 train / 872 validation, batch size 32, seed 42, constant LR). AdamW (lr 2e-5 everywhere) reached final/best accuracy 90.83% (loss 0.26245, epoch 3), with validation loss staying narrow across epochs. Muon (lr 0.003 on the 37 hidden weight matrices, AdamW lr 2e-5 fallback elsewhere) reached final accuracy 84.40% (loss 0.39393, epoch 3), best at epoch 1 (87.04%, loss 0.36956); its validation loss rose to 0.5706 at epoch 2 even as training loss kept falling, then eased at epoch 3 — a constant, undecayed learning rate may make mid-training validation loss more sensitive to a fixed step size, though a single run cannot confirm this. By final/best accuracy, AdamW performed better here. Muon's matrix LR (0.003) came from a predeclared stability check over {0.001, 0.003, 0.01} that preferred it despite 0.01 scoring better; AdamW's 2e-5 followed common practice, so the two were not comparably tuned. A separate matched 32-step diagnostic (see figure below; not a reconstruction of the full run) shows Muon's matrix-group update norm rising to plateau near 0.49 while AdamW's decays from 0.13 toward 0.05; gradient norms overlap for both, so the gap tracks the update rule and LR, not gradient magnitude.

2. Which solution appears flatter? We perturbed each best checkpoint on 256 shared validation examples along 8 paired random directions per tensor, scaled to epsilon times each tensor's own norm, and measured the mean loss increase — a curvature proxy: a flatter minimum tolerates larger perturbations with less loss increase. At epsilon=0.1, AdamW's mean increase was 0.00741 versus 0.00435 for Muon, suggesting Muon's checkpoint is locally flatter. But the paired Muon-minus-AdamW 95% interval is [-0.00976, 0.00365] — it includes zero, so the difference is not distinguishable from noise here. The checkpoints are also from different epochs (AdamW 3, Muon 1) with different baseline losses (0.2253 vs 0.3199). Eight random directions do not identify worst-case curvature, and the per-tensor scaling is not fully reparameterization-invariant, so these numbers describe sensitivity along sampled directions only, not a worst-case bound. A second estimate, the top Hessian eigenvalue via power iteration on Hessian-vector products with Hotelling deflation (to avoid mistaking largest-*magnitude* for largest-positive), reverses this ranking: 126.19 for AdamW versus 492.99 for Muon, whose dominant direction is in fact strongly negative (-1180.66); 492.99 is verified as a genuine, reproducible positive eigenvalue found by deflating that mode away, though not a proven global maximum (`SUPPLEMENTARY\_ANALYSIS.md`). Random directions mostly probe average-case curvature in this 67M-parameter space, while the top eigenvalue is the single sharpest direction a random probe is unlikely to find; neither measure is more "correct."

3. How reliable are these conclusions? Limited. Single seed per optimizer, so no run-to-run variance estimate. Historical per-step norms were never recorded; the 32-step diagnostic is a separate, short, matched proxy, not the full trajectory. Both use weight_decay=0.01 as decoupled decay, but Muon's matrix lr (0.003) makes its realized per-step decay ~150x AdamW's, confounding update-rule and regularization effects. Muon's full run took 3621s vs AdamW's 1603s (~2.26x) on this M4/MPS machine with an unfused implementation — hardware/implementation-specific, not a general claim (details in `SUPPLEMENTARY\_ANALYSIS.md`).

4. What would you change in a larger study? Use multiple seeds to quantify variance. Record real per-step norms during the full runs. Evaluate both sharpness measures at matched checkpoint epochs and estimate more than one eigenvalue. Tune AdamW's rate with the same predeclared procedure used for Muon and match effective weight decay, not the nominal coefficient. Time both optimizers with a fused Muon implementation first.



32-step matched diagnostic (full-resolution figure: deliverables/optimizer_diagnostics.png)

Sources: saved runs and CSV tables in deliverables/; figures provided separately.